



63,000-year-old human teeth

Ancient human teeth have been discovered in a small excavated cave in an island in Indonesia. <http://digit.in/oldTooth>



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Up, up and Away!

S

cience fiction is a core part of being a geek. Being a hard-core Star Wars fan, or a die hard Star Trek

fan, you would know very well that most interplanetary sci-fi stories come with their own armada of awe-inspiring spacecrafts. One of the most interesting things about these spacecrafts is what they run on. It is undeniable that these crafts would be – literally and figuratively – a shell of themselves without their ‘drives’ (which is what are going to call their major operational mechanisms from here on). Despite the wide variety of options from all of science fiction, these drives can be majorly classified into several broad types.

ION DRIVE ENGINES

Perhaps the most realistic option in sci-fi, ion-drive engines offer the direct advantage of reducing the amount of fuel needed for the same thrust, thus offering greater fuel efficiency. But how does it work?

Briefly speaking, ions are charged particles of any element. And since they carry a charge, their velocity can be manipulated with electricity. Putting a magnetic field in place gives an ion-drive the ability to accelerate ions to enormous velocities. These ions are then released to provide thrust to the craft. While the acceleration is very low due to the low mass involved in the thrust, this acceleration can be maintained for far longer than what is possible with chemical engines, making it ideal for space faring people.

Have you ever wondered what powers your favourite sci-fi spacecrafts?

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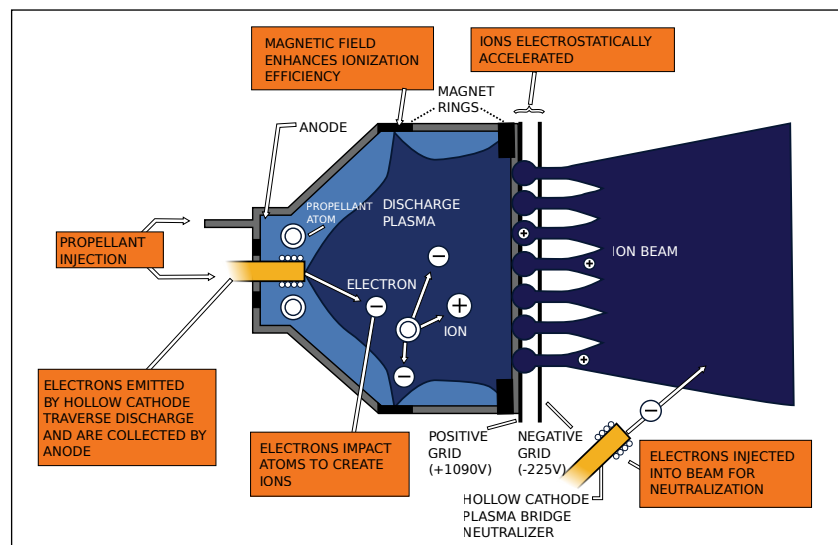
***Interesting fact:** Ion drive engines are real – to some extent. A number of deep space missions – current and planned – are using ion engines for propulsion.*

In terms of popularity, the Star Wars universe has seen a large number of these drives. In fact, the popular TIE fighters expand to Twin Ion Engine fighters. Although, popular franchise Battlestar Galactica often incorrectly refers to ion-drive powered ships achieving light speed, which is nowhere near possible.

DARK MATTER ENGINES

Dark matter has largely been an unexplainable mystery, be it for scientists in real life or fictional worlds. It has been used as the source or explanation for everything from super-powerful guns, a city full of metahumans, and super-powerful ships that can achieve unexplainable feats.

One of the most recurrent and popular cases is Futurama, where dark matter forms the primary fuel for starships. In the series, the only known source of dark matter is the excretion of Nibblonians, a species that existed even before the Big Bang (an interesting source, nonetheless). In the series, the dark matter is ‘burned’ in large furnaces, channeling it through an afterburner which gives the ship enough speed to cover the entire universe in a matter of days. But how is ‘faster than



How an ion engine works